

An Exact Boundary and Skin Atlas for the Finite Hatano–Nelson Chain

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Abstract

We solve the finite asymmetric nearest-neighbor chain across its full hopping cone. Positive open-boundary hopping is diagonally similar to a symmetric path matrix, yielding the exact Chebyshev spectrum.

1 Model and the algebraic core

For $N \geq 2$, let the open-boundary matrix $H_N = H_N(t_R, t_L)$ have $(H_N)_{j,j+1} = t_R$ and $(H_N)_{j+1,j} = t_L$, where $t_R, t_L \geq 0$. Evolution means $i\psi = H_N\psi$. When both hoppings are positive, put

$$g = \sqrt{t_R t_L}, \quad q = \sqrt{t_L/t_R}, \quad D = \text{diag}(1, q, \dots, q^{N-1}),$$

and let A_N be path adjacency. Define

$$\theta_m = \frac{m\pi}{N+1}, \quad S_{jm} = \sqrt{\frac{2}{N+1}} \sin(j\theta_m), \quad 1 \leq j, m \leq N.$$

Theorem 1 (finite-chain atlas). *For $t_R, t_L > 0$,*

$$D^{-1}H_N D = gA_N, \quad \det(zI - H_N) = g^N U_N\left(\frac{z}{2g}\right), \quad E_m = 2g \cos \theta_m. \quad (1)$$

The E_m are real and simple.

Proof. Since $D_j^{-1}t_R D_{j+1} = t_R q = g$ and $D_{j+1}^{-1}t_L D_j = t_L/q = g$, diagonal conjugation gives the first identity. Laplace expansion produces the continuant

$$P_0 = 1, \quad P_1 = z, \quad P_N = zP_{N-1} - t_R t_L P_{N-2}.$$

The recurrence for U_N gives (1). The discrete sine transform is orthogonal and diagonalizes A_N , which proves simplicity and reality. $\square$

References

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- [2] N. Hatano and D. R. Nelson, “Non-Hermitian delocalization and eigenfunctions,” *Phys. Rev. B* **58** (1998), 8384–8390. doi:10.1103/PhysRevB.58.8384.